

```
MariaDB [(none)]> CREATE DATABASE Kekeli_Corporation;  
Query OK, 1 row affected (0.006 sec)
```

```
MariaDB [(none)]> USE Kekeli_Corporation;  
Database changed
```

```
MariaDB [Kekeli_Corporation]> CREATE TABLE Employee(  
->     empNo VARCHAR(10) PRIMARY KEY,  
->     fName VARCHAR(30),  
->     lName VARCHAR(50),  
->     address VARCHAR(100),  
->     DOB DATE,  
->     sex VARCHAR(1) CHECK(sex IN('M','F')),  
->     position VARCHAR(30) CHECK(position IN('Manager','Team  
Leader','Analyst','Software Developer')),  
->     deptNo VARCHAR(10)  
-> );  
Query OK, 0 rows affected (0.023 sec)
```

```
MariaDB [Kekeli_Corporation]> CREATE TABLE Department(  
->     deptNo VARCHAR(10) PRIMARY KEY,  
->     deptName VARCHAR(70) NOT NULL,  
->     mgrEmpNo VARCHAR(10),  
->     FOREIGN KEY (mgrEmpNo) REFERENCES Employee(empNo)  
-> );  
Query OK, 0 rows affected (0.021 sec)
```

```
MariaDB [Kekeli_Corporation]> ALTER TABLE Employee  
-> ADD CONSTRAINT fk  
-> FOREIGN KEY (deptNo) REFERENCES Department(deptNo);  
Query OK, 0 rows affected (0.074 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [Kekeli_Corporation]> CREATE TABLE Project(  
->     projNo VARCHAR(10) PRIMARY KEY,  
->     projName VARCHAR(70),  
->     deptNo VARCHAR(10),  
->     FOREIGN KEY (deptNo) REFERENCES Department(deptNo)  
-> );  
Query OK, 0 rows affected (0.022 sec)
```

```
MariaDB [Kekeli_Corporation]> CREATE TABLE WorksOn(  
->     empNo VARCHAR(10),  
->     projNo VARCHAR(10),  
->     dateWorked DATE,  
->     hoursWorked INTEGER CHECK(hoursWorked BETWEEN 0 AND 40),  
->     PRIMARY KEY (empNo, projNo, dateWorked),  
->     FOREIGN KEY (empNo) REFERENCES Employee(empNo),  
->     FOREIGN KEY (projNo) REFERENCES Project(projNo)
```

```
-> );  
Query OK, 0 rows affected (0.024 sec)
```

```
MariaDB [Kekeli_Corporation]> INSERT INTO Department (deptNo, deptName) VALUES  
-> ('D1', 'Tech Innovation Hub'),  
-> ('D2', 'Agri-Tech Solutions'),  
-> ('D3', 'FinTech Infrastructure'),  
-> ('D4', 'GIS & Remote Sensing Division'),  
-> ('D5', 'Human Capital & Operations');  
Query OK, 5 rows affected (0.011 sec)
```

```
MariaDB [Kekeli_Corporation]> INSERT INTO Employee (empNo, fName, lName, address,  
DOB, sex, position, deptNo) VALUES  
-> ('E1', 'Kofi', 'Mensah', 'East Legon, Accra', '1985-05-12', 'M', 'Manager',  
'D1'),  
-> ('E2', 'Ama', 'Serwaa', 'Dansoman, Accra', '1990-08-22', 'F', 'Manager',  
'D2'),  
-> ('E3', 'Kwame', 'Osei', 'Santasi, Kumasi', '1988-11-03', 'M', 'Manager',  
'D3'),  
-> ('E4', 'Abena', 'Agyapong', 'Ayigya, Kumasi', '1992-02-14', 'F', 'Manager',  
'D4'),  
-> ('E5', 'Yaw', 'Boakye', 'Tarkwa, Western', '1987-09-30', 'M', 'Manager',  
'D5'),  
-> ('E6', 'Esi', 'Appiah', 'Madina, Accra', '1995-04-10', 'F', 'Software  
Developer', 'D1'),  
-> ('E7', 'Seth', 'Quartey', 'Osu, Accra', '1993-06-25', 'M', 'Analyst', 'D2'),  
-> ('E8', 'Adwoa', 'Darko', 'Taifa, Accra', '1996-12-01', 'F', 'Software  
Developer', 'D3'),  
-> ('E9', 'Kojo', 'Baah', 'Kwabenya, Accra', '1994-01-20', 'M', 'Team Leader',  
'D4'),  
-> ('E10', 'Efua', 'Tetteh', 'Spintex, Accra', '1997-07-15', 'F', 'Analyst',  
'D5');  
Query OK, 10 rows affected (0.008 sec)  
Records: 10 Duplicates: 0 Warnings: 0
```

```
MariaDB [Kekeli_Corporation]> UPDATE Department SET mgrEmpNo = 'E1' WHERE deptNo =  
'D1';  
Query OK, 1 row affected (0.006 sec)  
Rows matched: 1 Changed: 1 Warnings: 0
```

```
MariaDB [Kekeli_Corporation]> UPDATE Department SET mgrEmpNo = 'E2' WHERE deptNo =  
'D2';  
Query OK, 1 row affected (0.005 sec)  
Rows matched: 1 Changed: 1 Warnings: 0
```

```
MariaDB [Kekeli_Corporation]> UPDATE Department SET mgrEmpNo = 'E3' WHERE deptNo =  
'D3';  
Query OK, 1 row affected (0.004 sec)  
Rows matched: 1 Changed: 1 Warnings: 0
```

```
MariaDB [Kekeli_Corporation]> UPDATE Department SET mgrEmpNo = 'E4' WHERE deptNo = 'D4';
```

```
Query OK, 1 row affected (0.004 sec)
```

```
Rows matched: 1  Changed: 1  Warnings: 0
```

```
MariaDB [Kekeli_Corporation]> UPDATE Department SET mgrEmpNo = 'E5' WHERE deptNo = 'D5';
```

```
Query OK, 1 row affected (0.005 sec)
```

```
Rows matched: 1  Changed: 1  Warnings: 0
```

```
MariaDB [Kekeli_Corporation]> INSERT INTO Project (projNo, projName, deptNo) VALUES
```

```
-> ('GDS', 'Galamsey Detection System', 'D4'),  
-> ('MMAR', 'Mobile Money API Refactor', 'D3'),  
-> ('CYP', 'Cocoa Yield Predictor', 'D2'),  
-> ('ACPU', 'Ashesi Campus Portal UI', 'D1'),  
-> ('SPA', 'Staff Payroll Automation', 'D5');
```

```
Query OK, 5 rows affected (0.016 sec)
```

```
Records: 5  Duplicates: 0  Warnings: 0
```

```
MariaDB [Kekeli_Corporation]> INSERT INTO WorksOn (empNo, projNo, dateWorked, hoursWorked) VALUES
```

```
-> ('E6', 'ACPU', '2026-06-20', 8),  
-> ('E9', 'GDS', '2026-06-21', 6),  
-> ('E8', 'MMAR', '2026-06-21', 7),  
-> ('E7', 'CYP', '2026-06-22', 5),  
-> ('E10', 'SPA', '2026-06-22', 4),  
-> ('E1', 'ACPU', '2026-06-23', 2),  
-> ('E2', 'CYP', '2026-06-23', 3),  
-> ('E3', 'MMAR', '2026-06-24', 4),  
-> ('E4', 'GDS', '2026-06-24', 5),  
-> ('E5', 'SPA', '2026-06-24', 2);
```

```
Query OK, 10 rows affected (0.006 sec)
```

```
Records: 10  Duplicates: 0  Warnings: 0
```

```
MariaDB [Kekeli_Corporation]> CREATE VIEW FemaleManagedProjects AS
```

```
-> SELECT p.projNo,p.projName  
-> FROM Project p, Employee e  
-> WHERE p.deptNo=e.deptNo AND (e.sex='F' AND e.position='Manager')  
-> ORDER BY p.projNo ASC;
```

```
Query OK, 0 rows affected (0.015 sec)
```

```
MariaDB [Kekeli_Corporation]> select * from FemaleManagedProjects;
```

```
+-----+-----+  
| projNo | projName                |  
+-----+-----+  
| CYP    | Cocoa Yield Predictor    |  
| GDS    | Galamsey Detection System |  
+-----+-----+
```

```
2 rows in set (0.005 sec)
```

```

MariaDB [Kekeli_Corporation]> CREATE VIEW EmployeeSummary AS
-> SELECT e.empNo,e.fName,e.lName,p.projName,w.hoursWorked
-> FROM Employee e, Project p, WorksOn w
-> WHERE e.empNo=w.empNo AND p.projNo=w.projNo;
Query OK, 0 rows affected (0.016 sec)

```

```

MariaDB [Kekeli_Corporation]> select * from EmployeeSummary;

```

empNo	fName	lName	projName	hoursWorked
E1	Kofi	Mensah	Ashesi Campus Portal UI	2
E6	Esi	Appiah	Ashesi Campus Portal UI	8
E2	Ama	Serwaa	Cocoa Yield Predictor	3
E7	Seth	Quartey	Cocoa Yield Predictor	5
E4	Abena	Agyapong	Galamsey Detection System	5
E9	Kojo	Baah	Galamsey Detection System	6
E3	Kwame	Osei	Mobile Money API Refactor	4
E8	Adwoa	Darko	Mobile Money API Refactor	7
E10	Efua	Tetteh	Staff Payroll Automation	4
E5	Yaw	Boakye	Staff Payroll Automation	2

```

10 rows in set (0.004 sec)

```

```

MariaDB [Kekeli_Corporation]> CREATE VIEW EmpProject (empNo, projNo, totalHours)
-> AS SELECT w.empNo, w.projNo, SUM(hoursWorked)
-> FROM Employee e, Project p, WorksOn w
-> WHERE e.empNo = w.empNo AND p.projNo = w.projNo
-> GROUP BY w.empNo, w.projNo;
Query OK, 0 rows affected (0.014 sec)

```

```

MariaDB [Kekeli_Corporation]> SELECT *
-> FROM EmpProject;

```

empNo	projNo	totalHours
E1	ACPU	2
E10	SPA	4
E2	CYP	3
E3	MMAR	4
E4	GDS	5
E5	SPA	2
E6	ACPU	8
E7	CYP	5
E8	MMAR	7
E9	GDS	6

```

10 rows in set (0.006 sec)

```

```

MariaDB [Kekeli_Corporation]> SELECT projNo

```

```
-> FROM EmpProject
-> WHERE projNo='SCCS';
Empty set (0.007 sec)
```

```
MariaDB [Kekeli_Corporation]> SELECT COUNT(projNo)
-> FROM EmpProject
-> WHERE empNo='E1';
```

```
+-----+
| COUNT(projNo) |
+-----+
|              1 |
+-----+
```

1 row in set (0.007 sec)

```
MariaDB [Kekeli_Corporation]> SELECT empNo, totalHours
-> FROM EmpProject
-> GROUP BY empNo;
```

```
+-----+-----+
| empNo | totalHours |
+-----+-----+
| E1    |          2 |
| E10   |          4 |
| E2    |          3 |
| E3    |          4 |
| E4    |          5 |
| E5    |          2 |
| E6    |          8 |
| E7    |          5 |
| E8    |          7 |
| E9    |          6 |
+-----+-----+
```

10 rows in set (0.004 sec)